

■ EDA & DATA CLEANING Interview Questions & Answers

For Freshers — Simple English, Easy to Remember

50+ Most-Asked EDA & Data Cleaning Questions with Clear Answers

SECTION 1 — What is EDA?

Q1. What is EDA (Exploratory Data Analysis)?

Answer: EDA means exploring your data BEFORE doing any analysis or model. It's like reading a book's summary before reading the full book. You look at the data's shape, find patterns, check for problems (missing values, wrong data), and understand what the data is telling you.

Example: Think of it as 'getting to know your data' step.

Q2. Why is EDA important?

Answer: EDA helps you: (1) Understand what data you have. (2) Find missing values or errors. (3) Find outliers (unusual values). (4) Understand relationships between columns. (5) Decide what cleaning is needed. (6) Get ideas for new features. Without EDA, you might build wrong models or draw wrong conclusions.

Remember: Always do EDA first — never skip it!

Q3. What are the steps of EDA?

Answer: Step 1: Load data and check shape (rows, columns). Step 2: Check data types and column names. Step 3: Check missing values. Step 4: Check basic statistics (mean, min, max). Step 5: Look at distributions (histograms). Step 6: Find relationships (correlation, scatter plots). Step 7: Find outliers. Step 8: Summarise findings.

Q4. What does `df.describe()` show you?

Answer: `df.describe()` shows statistics for all number columns: count (how many values), mean (average), std (standard deviation), min, 25% (Q1), 50% (median), 75% (Q3), max. It gives you a quick summary of the data.

Q5. What is the difference between Univariate, Bivariate, and Multivariate analysis?

Answer: Univariate = analysing ONE column at a time. Example: distribution of ages. Bivariate = analysing TWO columns together. Example: salary vs experience. Multivariate = analysing THREE or more columns. Example: correlation matrix of all columns.

Remember: Uni=1, Bi=2, Multi=many

Q6. What is a distribution? What does it tell you?

Answer: A distribution shows how values are spread — how many times each value (or range) appears. A histogram shows this. Looking at distribution tells you: Is data symmetric? Is it skewed? Are there outliers? What is the most common value?

Example: Age distribution of customers — most between 25-40 with a few very old customers.

Q7. What is skewness in simple words?

Answer: Skewness tells you if your data is balanced or leans to one side. Symmetric (skewness=0): data is balanced like a bell. Right skew (positive): most data is on the left, tail goes to the right. Example: income (few very rich people). Left skew (negative): most data is on the right, tail goes to the left.

Remember: Right skew = tail on right = few very high values pulling the average up.

Q8. What is correlation? How do you read a correlation matrix?

Answer: Correlation measures how strongly two columns move together. Value between -1 and +1. +1 = perfect positive (one goes up, other goes up). -1 = perfect negative (one goes up, other goes down). 0 = no relationship. In a heatmap, dark red/blue = strong, white/light = weak.

Example: Height and weight usually have correlation of 0.7+ (positive).

Q9. What is the difference between correlation and causation?

Answer: Correlation = two things move together. Causation = one thing CAUSES the other. Ice cream sales and drowning accidents are correlated (both increase in summer) — but ice cream doesn't CAUSE drowning. Summer (heat) causes both.

Remember: Correlation does NOT mean one causes the other — very important concept!

SECTION 2 — Missing Values

Q10. What are missing values? Why do they occur?

Answer: Missing values are empty cells — no data recorded for that row+column. They occur because: user skipped a form field, data entry error, system failure, data not collected, or genuinely not applicable.

Example: Age column has NaN for some rows — maybe the user didn't fill it in.

Q11. How do you find missing values in Python?

Answer: `df.isnull()` — returns True/False for each cell. `df.isnull().sum()` — count of missing values per column. `df.isnull().sum()/len(df)*100` — percentage missing. `df.isnull().any()` — which columns have at least one missing value.

Q12. What are the three types of missing data? (Simple version)

Answer: MCAR (Missing Completely At Random) — missing by accident, no pattern. Example: random survey entries left blank. MAR (Missing At Random) — missing based on other columns. Example: older people skip income question. MNAR (Missing Not At Random) — missing because of the value itself. Example: very high earners hide income.

Remember: MCAR = safe to delete. MAR = can impute. MNAR = hardest, needs careful handling.

Q13. What are the ways to handle missing values?

Answer: 1) Delete rows: `df.dropna()` — only if very few rows are missing (< 5%). 2) Fill with mean: `df['col'].fillna(df['col'].mean())` — for symmetric numerical data. 3) Fill with median: for skewed data or data with outliers. 4) Fill with mode: for categorical columns (most common value). 5) Fill with a constant: `df.fillna(0)` or `df.fillna('Unknown')`. 6) Forward fill: `df.ffill()` — use previous row's value (good for time series).

Remember: Never just delete all missing data without thinking — you may lose important information.

Q14. When should you NOT delete rows with missing values?

Answer: When too many rows are missing (> 10-20%), deleting creates bias. When the missing pattern is not random (MNAR). When the dataset is already small and you can't afford to lose rows.

SECTION 3 — Outliers

Q15. What is an outlier?

Answer: An outlier is a value that is very different from the rest of the data. Example: if most salaries are 30,000-80,000 and one is 5,000,000 — that's an outlier. Outliers can be data errors OR genuine extreme values.

Q16. How do you detect outliers?

Answer: 1) Box plot — dots beyond whiskers are outliers. 2) IQR Method: Calculate Q1, Q3. $IQR = Q3 - Q1$. Lower bound = $Q1 - 1.5 * IQR$. Upper bound = $Q3 + 1.5 * IQR$. Values outside these bounds = outliers. 3) Z-score: If z-score > 3 or < -3 , it's an outlier. 4) Scatter plot — points far from the main cluster.

Q17. What is IQR? How do you use it to find outliers?

Answer: IQR = Interquartile Range = $Q3 - Q1$. $Q1$ = 25th percentile (bottom 25% of data). $Q3$ = 75th percentile (top 25% of data). Outlier if value $< Q1 - 1.5 * IQR$ or value $> Q3 + 1.5 * IQR$.

Example: If $Q1=20$, $Q3=60$, $IQR=40$. Lower = $20-60=-40$. Upper = $60+60=120$. Any value below -40 or above 120 is outlier.

Q18. How do you handle outliers?

Answer: 1) Remove them — if they are data errors (typos, wrong entries). 2) Cap/Winsorise — replace extreme values with the boundary value (e.g., replace anything > 100000 with 100000). 3) Transform data — use log transformation to reduce the effect. 4) Keep them — if they are real and meaningful (CEO salary is genuinely high). 5) Analyse separately — sometimes outliers tell the most interesting story.

Remember: First understand WHY it's an outlier, then decide what to do.

Q19. What is the difference between an outlier and a wrong value?

Answer: An outlier is an extreme but possibly real value. A wrong value is a data entry error. Example: age = 150 is a WRONG value (impossible). Age = 95 might be an outlier but is possible. Always validate with domain knowledge.

SECTION 4 — Data Cleaning Techniques

Q20. What is Data Cleaning? Why is it needed?

Answer: Data cleaning is the process of fixing or removing wrong, incomplete, duplicate, or irrelevant data. Real-world data is always messy. If you analyse dirty data, you get wrong results — 'Garbage in, Garbage out'. Typically 60-80% of a data analyst's time is spent cleaning data.

Q21. How do you handle duplicate rows?

Answer: Find duplicates: `df.duplicated()` — returns True for duplicate rows. Count: `df.duplicated().sum()`. Remove: `df.drop_duplicates()`. Remove based on specific columns: `df.drop_duplicates(subset=['email'])`.

Q22. How do you fix inconsistent text data?

Answer: Problems: 'Mumbai', 'mumbai', 'MUMBAI', ' Mumbai ' (extra space) all mean the same but Python sees them differently. Fix: `df['city'].str.lower()` — all lowercase. `df['city'].str.strip()` — remove spaces. `df['city'].str.title()` — Title Case. Then use `replace()` for any remaining inconsistencies.

Example: `df['gender'] = df['gender'].str.lower().replace({'m':'male','f':'female'})`

Q23. How do you convert data types?

Answer: Common issues: dates stored as strings, prices stored as strings with currency symbol (Rs.500). Fix: `pd.to_datetime(df['date'])` — convert to date. `pd.to_numeric(df['price'], errors='coerce')` — convert to number. `df['age'].astype(int)` — convert to integer.

Example: `df['salary'] = df['salary'].str.replace('Rs.', '').str.replace(',', '.').astype(float)`

Q24. How do you handle wrong data formats?

Answer: Date formats: 2024-01-15, 15/01/2024, Jan 15 2024 — all different formats. Use `pd.to_datetime(df['date'], format='%d/%m/%Y')` to specify the format. For phone numbers, emails — use string cleaning and regex validation.

Q25. What is normalisation and standardisation?

Answer: Both scale numerical data to a similar range so no one column dominates. Normalisation (Min-Max Scaling): rescales values to 0-1 range. Use when you need bounded values. Standardisation (Z-score): rescales to mean=0, std=1. Use when data needs to follow normal distribution. In Python: from `sklearn.preprocessing` import `MinMaxScaler`, `StandardScaler`.

Example: Salary (50000) and Age (25) are very different scales — scaling makes them comparable.

Q26. What is feature engineering? Give examples.

Answer: Feature engineering means creating new useful columns from existing ones to give more information to your analysis or model. Examples: From a date column, extract Day, Month, Year, Day of Week. From `FirstName` + `LastName`, create `FullName`. From `OrderDate` and `DeliveryDate`, create `DeliveryDays`. Create age groups from exact age (Child/Teen/Adult/Senior).

Remember: New features often improve analysis significantly.

Q27. What is One-Hot Encoding? When do you use it?

Answer: One-Hot Encoding converts categorical (text) columns into numbers by creating a new column for each category with 0 or 1 values. Use for columns with no natural order (like City, Gender, Color). In Python: `pd.get_dummies(df['city'])`.

Example: City column with Mumbai/Delhi/Pune becomes 3 columns: is_Mumbai(0/1), is_Delhi(0/1), is_Pune(0/1).

Remember: Use One-Hot for no-order categories. Use Label Encoding (0,1,2...) for ordered ones.

Q28. What is Label Encoding?

Answer: Label Encoding converts categories to numbers: 0, 1, 2, 3... Use for columns that have a natural order (like ratings: Bad=0, Average=1, Good=2, Excellent=3).

Example: Size column: Small=0, Medium=1, Large=2, XL=3.

Q29. How do you check data quality?

Answer: Check: (1) Completeness — any missing values? (2) Uniqueness — any duplicates? (3) Consistency — same format throughout? (4) Validity — values in expected range? (age > 0, salary > 0). (5) Accuracy — does it match real world? (6) Timeliness — is data up to date?

Q30. What is a data type mismatch? How do you fix it?

Answer: Data type mismatch = a column has wrong data type. Example: 'Age' column has values like '25', '30abc', 'N/A' stored as strings instead of integers. Fix: use `pd.to_numeric(df['Age'], errors='coerce')` — this converts valid numbers and puts NaN for invalid ones.

SECTION 5 — Statistics for Data Analysis

Q31. What is Mean, Median, Mode?

Answer: Mean = average (sum of values / count). Best for symmetric data. Median = middle value when sorted. Best for skewed data or data with outliers. Mode = most frequently occurring value. Best for categorical data.

Example: Salaries: [20K, 25K, 30K, 30K, 1000K]. Mean=221K (misleading!). Median=30K (better). Mode=30K.

Remember: For income/house prices, always use Median — outliers skew the Mean heavily.

Q32. What is Standard Deviation?

Answer: Standard Deviation (std) tells you how spread out the data is. Low std = values are close to the mean. High std = values are spread far from the mean.

Example: Class A test scores: 70,71,69,70,71 — std is very low (consistent). Class B: 40,95,30,85,100 — std is very high (inconsistent).

Q33. What is a percentile?

Answer: A percentile tells you what percentage of data falls below a value. 25th percentile (Q1) = 25% of values are below this. 50th percentile = median. 75th percentile (Q3) = 75% of values are below this. 90th percentile salary = only 10% earn more than this.

Q34. What is the difference between variance and standard deviation?

Answer: Variance = average of squared differences from the mean. Standard Deviation = square root of variance. Standard deviation is in the same unit as the data (easier to understand). Both measure spread — but use std in reports because it's in the original unit.

Q35. What is a normal distribution?

Answer: Normal distribution is the famous 'bell curve' shape. Most values are near the mean. Values become less frequent as you go to extremes. Many real-world things follow normal distribution: height, test scores, weight. Important because many statistical methods assume data is normally distributed.

EDA Checklist — Use This in Every Project

Step	Command/Action	What to Look For
1. Load Data	<code>pd.read_csv('file.csv')</code>	File loads without errors?
2. Check Shape	<code>df.shape</code>	How many rows and columns?
3. Preview	<code>df.head()</code> , <code>df.tail()</code>	Does data look right?
4. Data Types	<code>df.info()</code>	Are types correct? Dates as dates?
5. Basic Stats	<code>df.describe()</code>	Min/Max reasonable? Outliers?
6. Missing Values	<code>df.isnull().sum()</code>	Which columns have NaN?
7. Duplicates	<code>df.duplicated().sum()</code>	Any duplicate rows?
8. Unique Values	<code>df['col'].value_counts()</code>	Inconsistent categories?
9. Distribution	<code>df.hist()</code> / <code>sns.histplot()</code>	Skewed? Normal? Outliers?
10. Correlation	<code>df.corr()</code> / <code>sns.heatmap()</code>	Strong relationships?
11. Outliers	<code>sns.boxplot()</code>	Any extreme values?
12. Clean	<code>dropna</code> / <code>fillna</code> / <code>replace</code> / <code>astype</code>	Fix all found issues

Data cleaning is 80% of the job — master it and you're ahead of most candidates! ■